SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – C Code Output Prediction

Course Code:	CSA0309	**Course Title:**	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	**Assessment:**	Assessment Tool 1 – C Code Output Prediction
Weightage:	35% of CIA	**Total Marks:**	100
CO1 Statement:	*Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.*		
Student Name:	KARTHI M E	**Reg. No.:**	192521175
Section / Batch:	SLOT-B	**Date:**	08/07/2026

Code of Conduct

I am Karthi M E (192521175) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: KARTHI M E

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1 - 2	20
Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure	Classifications of Linear data structure	3 - 4	20
Search Data Structure	Linear Search and Binary Search	5 -6	20
Recursion, Performance analysis	Recursion	7 - 8	20
Array Data Structures	Implementations Using Array data structure	9 - 10	20
		GRAND TOTAL:	**100 Marks**

SIMATS Online Lab

⚠ Not secure simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=340

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

KARTHIK M E
192521175RB

CO1_AT1_C CODE OUTPUT PREDICTION

← Back

QUESTIONS

Q1
C Code – Pointer with Array
10 marks

Q2
C Code - Linked List Node
10 marks

Q3
C Code - Binary Search Complexity
10 marks

Q4
C Code - Pointer Increment
10 marks

Q5
C Code - Array Address
10 marks

Q2 C Code - Linked List Node

10 marks

```
#include<stdio.h>
struct Node{
int data;
struct Node *next;
};
int main(){
struct Node n1,n2;
n1.data=10;
n2.data=20;
n1.next=&n2;
n2.next=NULL;
printf("%d",n1.next->data);
}
```

Your Answer

SIMATS Online Lab

⚠ Not secure simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=340

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

KARTHIK M E
192521175RB

CO1_AT1_C CODE OUTPUT PREDICTION

← Back

QUESTIONS

Q1
C Code – Pointer with Array
10 marks

Q2
C Code - Linked List Node
10 marks

Q3
C Code - Binary Search Complexity
10 marks

Q4
C Code - Pointer Increment
10 marks

Q5
C Code - Array Address
10 marks

Q3 C Code - Binary Search Complexity

10 marks

```
while(low<=high){
mid=(low+high)/2;
}
```

Your Answer

Binary search

Save Answer

SIMATS Online Lab

Not secure simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=340

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

KARTHIK M E
192521175RB

CO1_AT1_C CODE OUTPUT PREDICTION

Back

QUESTIONS

Q1
C Code – Pointer with Array
10 marks

Q2
C Code - Linked List Node
10 marks

Q3
C Code - Binary Search Complexity
10 marks

Q4
C Code - Pointer Increment
10 marks

Q5
C Code - Array Address
10 marks

Q5 C Code - Array Address

10 marks

```
int arr[]={5,10,15};
printf("%d",*(arr+1));
```

Your Answer

10

Save Answer

SIMATS Online Lab

Not secure simatslab.saveetha.com/practice_app/classactivity/attempt?assignment_id=340

SIMATS ENGINEERING

SIMATS Online Programming Lab

DS PROGRAMMING

KARTHIK M E
192521175RB

CO1_AT1_C CODE OUTPUT PREDICTION

Back

QUESTIONS

Q1
C Code – Pointer with Array
10 marks

Q2
C Code - Linked List Node
10 marks

Q3
C Code - Binary Search Complexity
10 marks

Q4
C Code - Pointer Increment
10 marks

Q5
C Code - Array Address
10 marks

Q4 C Code - Pointer Increment

10 marks

```
#include<stdio.h>
int main(){
int a[]={5,10,15};
int *p=a;
p++;
printf("%d",*p);
}
```

Your Answer

10